

Aegis Pipeline (v4)

Input
ERC-4337 account
contracts

VOIR Construction

Semantic lifting of validation logic

- Program representation extraction
- Custom semantic fact lifting
- Built-in ERC-4337 semantic profile

Output: VOIR graph and semantic facts

Semantic Checker Layer

Predicate-violation detection over VOIR

- Six VOIR semantic checkers
- Witness constraint construction

Output: violated predicates and witness constraints

Evidence Consolidation

Contract-level finding synthesis

- Source location projection
- Finding aggregation

Output: auditable findings

**Aegis
Analysis
Report**

type, trace,
locations

Underlined submodules are implemented with reused open-source components.